Eye Disease Classification Using Deep Learning

1. Project Overview

This project aims to classify eye diseases from retinal images using deep learning techniques.
Two approaches were implemented:

1. A custom CNN model built from scratch.
2. EfficientNetB0 pretrained model using Transfer Learning.

The goal was to compare learning features from the dataset directly versus using pretrained visual features.

2. Data Preparation

The images were resized and preprocessed before training. Data augmentation techniques were applied to improve generalization and reduce overfitting.

The dataset was imbalanced, where some disease classes contained more samples than others. The models were initially trained using the original dataset distribution.

3. CNN Model

A custom CNN architecture was developed using convolution layers, batch normalization, pooling layers, global average pooling, dense layers and dropout.

The model was trained using Adam optimizer and categorical cross entropy loss.

4. Pretrained Model

EfficientNetB0 pretrained on ImageNet was used for transfer learning. A new classification head was added and fine-tuning was applied to adapt the model for eye disease classification.

5. Handling Class Imbalance

Since the dataset was not balanced, different techniques were tested to reduce the effect of class imbalance.

Oversampling was applied to increase the number of samples in minority classes. Class weighting was also tested by assigning higher importance to minority classes during loss calculation.

However, both methods resulted in only a very small improvement in performance.

This indicates that class imbalance was not the main limitation affecting the models. The original dataset distribution was already sufficient for learning useful features.

Possible reasons:
- Eye disease classes have strong visual differences.

- Data augmentation increased training diversity.
- The CNN architecture was deep enough to extract important patterns.
- The remaining errors were related to similarity between some disease features rather than only class distribution.

6. Results Analysis

Both CNN and EfficientNetB0 achieved similar performance.

The small difference between the models can be explained because the custom CNN was able to learn meaningful retinal features directly from the images. Although pretrained models usually provide better results on small datasets, the dataset characteristics in this project allowed the CNN to achieve comparable performance.

7. Comparison

Custom CNN:
- Learns features from scratch.
- Requires more training.
- Achieved strong results due to effective preprocessing and architecture.

EfficientNetB0:
- Uses pretrained ImageNet features.
- Converges faster.
- Provides transfer learning benefits.

8. Conclusion

Deep learning models can effectively classify eye diseases from retinal images.

Experiments with oversampling and class weighting were performed to handle class imbalance, but the improvement was minimal. This shows that the original dataset distribution was adequate and that model architecture and feature learning had a greater impact on performance.

Future improvements include testing more architectures, increasing dataset size, and applying explainability methods such as Grad-CAM.